THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	$72 = 2^3 \times 3^2$; multiply by the primes with odd power
2	**(c)** 3^5	Add the exponents on top, then subtract the bottom exponent
3	**(d)** 22	Expand with powers of the base
4	**(d)** 75	$4x + 60 = 360$
5	**(a)** 2	Alternate digit sums must differ by 0 or 11
6	**(c)** $6x^2 + 19x + 10$	Multiply term by term and collect like terms
7	**(b)** 63	One part $= 9$
8	**(b)** 1200	CP = SP×100/(100+5)
9	**(c)** 24	Height splits the base: $h^2 = 30^2 - 18^2 = 576$
10	**(c)** 2:3	A:B:C $= 6 : 8 : 9$
11	**(c)** 8	$n = 6$, faces $= n + 2$
12	**(c)** 52	$(10 \times 40 + 15 \times 60) \div 25$
13	**(d)** $5x + 12$	$3x + 12 + 2x$
14	**(c)** 252 cm²	$300 - 48$
15	**(a)** 3	$192 = 2^6 \times 3$
16	**(a)** $\frac{9}{4}$	$(a/b)^{-n} = (b/a)^n$
17	**(d)** 101101	Repeated division by 2
18	**(d)** 20 cm	$2k(3 + 5) = 64$ gives $k = 4$
19	**(a)** 0	The number must be even and its digit sum a multiple of 3
20	**(c)** $4x^2 - 12xy + 9y^2$	$(a - b)^2 = a^2 - 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$2916 = 2^2 \times 3^6$ (1); $\sqrt{2916} = 54$ (1).	1+1
22	$(2^3)^3 = 2^9$, so $2^9 \div 2^5 = 2^4$ (1) $= 16$ (1).	1+1
23	$1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$ (1); $= 53$ (1).	1+1
24	One part $= 360 \div 18 = 20°$ (1); largest $= 160°$ (1).	1+1
25	8456 divided by 11 leaves remainder 8 (1); add $11 - 8 = 3$ (1).	1+1
26	$(2x)^2 + 2(2x)(5y) + (5y)^2$ (1) $= 4x^2 + 20xy + 25y^2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$1872 = 2^4 \times 3^2 \times 13$ (1); multiplier 13 (1); product 24336, root 156 (1).	1+1+1
28	$2^2 \times 3^2$ (1+1); value $4 \times 9 = 36$ (1).	1+1+1
29	$1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 45$ (1); $45 = 8 \times 5 + 5$; $5 = 8 \times 0 + 5$ (1); 55_8 (1).	1+1+1
30	$4x + 90 = 360$ (1); $x = 67$ (1); angles 67°, 82°, 97°, 112° (1).	1+1+1
31	LCM of 12 and 15 is 60 (1); first multiple not below 1000 (1); 1020 (1).	1+1+1
	OR: $6x^2 - 15x + 4x - 10$ (2); $= 6x^2 - 11x - 10$ (1).	2+1
32	Milk 35 L, water 5 L (1); new water $= \frac{3}{7} \times 35 = 15$ L (1); add 10 L (1).	1+1+1
	OR: After first discount: ₹ 3600 (1); SP = ₹ 3420 (1); equivalent discount = 14.5% (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Next perfect square above 1500 is $39^2 = 1521$ (2); students needed $= 21$ (2); rows $= 39$ (1).	2+2+1
34	Numerator power of 2: $2^3 \div 2^{-2} = 2^5$ (2); for 3: $3^{-1} \div 3^1 = 3^{-2} = \frac{1}{3^2}$ (1); result $\frac{2^5}{3^2}$ (1); value $\frac{32}{9}$ (1).	2+1+1+1
35	Add in binary: 11100_2 (3); decimal check $13 + 9 + 6 = 28$ (2).	3+2
36	Diagonals meet at right angles, so $\angle OAB + \angle OBA = 90^\circ$ (1); $2x + 10 + 3x - 5 = 90$ gives $x = 17$ (2); $\angle A = 2 \times 44 = 88^\circ$ (1); $\angle B = 2 \times 46 = 92^\circ$ (1).	1+2+1+1
	OR: Steps: 9541-1459=8082; 8820-0288=8532; 8532-2358=6174 (3); total 3 steps (2).	3+2
37	Plot $= (3x + 2)^2 = 9x^2 + 12x + 4$ (2); pond $= 4x^2$ (1); remaining $= 5x^2 + 12x + 4$ (1); at $x = 10$: 624 m^2 (1).	2+1+1+1
	OR: Total work $= 12 \times 15 = 180$ man-days (1); (i) $180 \div 18 = 10$ days (2); (ii) $180 \div 9 = 20$ men (2).	1+2+2
38	CP = 2100×100/105 = ₹ 2000 (2); new SP − CP = ₹ 400 (1); percent = 20% gain (2).	2+1+2
	OR: (i) $\sqrt{18^2 + 24^2} = 30$ m (2); (ii) saved $= 18 + 24 - 30 = 12$ m (1); (iii) total $= 18 + 24 + 30 = 72$ m (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 24	Subtract the added number
2	**(c)** 1800	CP = SP×100/(100−15)
3	**(d)** 84	Other side $= 24$; perimeter $= 2(18 + 24)$
4	**(a)** 24	Total work $= 12 \times 8 \times 15 = 1440$ man-hours
5	**(b)** 12	Base has $n = 6$ sides; edges $= 2n$
6	**(c)** 42	Sum changes by 6
7	**(a)** $x^2 + 7x + 10$	Multiply term by term
8	**(a)** 12	$b_1h_1 = b_2h_2$: $180/15$
9	**(c)** 9	Only the ones digit 7 matters: 7^2 ends in 9
10	**(b)** 7	$x + 4 = 11$
11	**(c)** CXI	44+67=111
12	**(b)** 34 cm	AO = OC; find x first, then AC = 2 AO
13	**(a)** 8262	Largest arrangement minus smallest arrangement
14	**(c)** 2600	$(63 + 37)(63 - 37) = 100 \times 26$
15	**(b)** 18 L	$\frac{3}{10} \times 60$
16	**(b)** 1800	MP = SP×100/(100−15)
17	**(d)** 48	Half of other diagonal $= \sqrt{30^2 - 18^2} = 24$
18	**(d)** 720	One part $= 1350 \div 15 = 90$
19	**(c)** 12	Each edge is shared by 2 faces: $E = 3F/2$
20	**(a)** 42.5	Arrange in ascending order first

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	One part $= 98 \div 14 = 7$ (1); numbers 35 and 63 (1).	1+1
22	CP = SP×100/(100+15) (1) = ₹ 1200 (1).	1+1
23	Height bisects the base, $h^2 = 30^2 - 18^2$ (1); $h = 24$ cm (1).	1+1
24	One part $= 143 \div 13 = 11$ (1); numbers 55 and 88 (1).	1+1
25	12 edges (1); sum $= 4(15 + 9 + 6) = 120$ cm (1).	1+1
26	Total marks $= 15 \times 60 + 10 \times 70 = 1600$ (1); mean $= 64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Initial foot distance $= \sqrt{25^2 - 24^2} = 7$ m (1); new height $= \sqrt{25^2 - 15^2} = 20$ m (1); slip $= 24 - 20 = 4$ m (1).	1+1+1
28	Total work $= 10 \times 6 \times 24 = 1440$ man-hours (1); workers' hours per day $= 12 \times 8 = 96$ (1); days $= 15$ (1).	1+1+1
29	(i) 8 (1); (ii) $(4 - 2)^3 = 8$ (1); (iii) $64 - 8 = 56$ (1).	1+1+1
30	Total weight $= 10 \times 42 + 15 \times 48 = 1140$ kg (2); mean $= 45.6$ kg (1).	2+1
31	Perimeter $9x + 9$ cm (2); for $x = 2$: $9 \times 2 + 9 = 27$ cm (1).	2+1
	OR: Inner rectangle $= (40 - 4) \times (30 - 4) = 936$ m² (2); path $= 1200 - 936 = 264$ m² (1).	2+1
32	$12000 = 2^5 \times 3 \times 5^3$ (1); divisor 12 (1); quotient 1000, root 10 (1).	1+1+1
	OR: $\frac{1}{8} + \frac{1}{9}$ (1); common denominator (1); sum $\frac{17}{72}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 6 × 5 = 30

Q.	Expected Answer / Value Points	Marks

33	Milk $= 24$ L, water $= 36$ L (2); new water $= 48$ L (1); water to add $= 12$ L (2).	2+1+2
34	Prism: $7, 10, 15$; pyramid: $6, 6, 10$ (2); differences: faces 1 (1), vertices 4 (1), edges 5 (1).	2+1+1+1
35	(i) changes: 4, 5, 3, -4, -5 (2); (ii) mean $= 24$ (2); (iii) largest change 5°C (1).	2+2+1
36	$x^2 + 8x + 15 - x^2 - 4x$ (2); $4x + 15$ (1); at $x = 2$: 23 (2).	2+1+2
	OR: Rectangle $= 200$ cm² (1); triangle $= \frac{1}{2} \times 20 \times 8 = 80$ cm² (2); total $= 280$ cm² (1); percentage $= 28.57\%$ (1).	1+2+1+1
37	$\sqrt[3]{1728} = 12$ cm (2); TSA $= 6s^2 = 6 \times 12^2 = 864$ cm² (3).	2+3
	OR: Number of cells $= \dfrac{8 \times 10^1}{2 \times 10^{-5}}$ (2); $= \frac{8}{2} \times 10^{1+5}$ (2); 4×10^6 (1).	2+2+1
38	Smallest $10000_2 = 16$ (2); largest $11111_2 = 31$ (2); count $32 - 16 = 16$ (1).	2+2+1
	OR: Diagonals of a rectangle are equal: $3x + 5 = 2x + 15$ gives $x = 10$ (2); AC = BD $= 35$ cm (1); OA $= 17.5$ cm (1); wire $= 2 \times 35 = 70$ cm (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 15	$3 \times 9 - 5 \times 4 + 8$
2	**(b)** 216 cm²	Other half-diagonal $= \sqrt{15^2 - 9^2} = 12$; area $= \frac{1}{2} \times 18 \times 24$
3	**(b)** 1.2	1.44 is the square of 1.2
4	**(d)** 4	$3x = 12$
5	**(c)** 7	It is $111_2 = 2^3 - 1$
6	**(a)** 17	Adjacent angles are supplementary
7	**(d)** 60	Find the LCM
8	**(c)** 100	$(7+3)^2$
9	**(a)** 15 hours	More pipes, less time
10	**(a)** 3000	P = SI×100/(R×T)
11	**(b)** 30	Height difference 18 m; $\sqrt{18^2 + 24^2} = 30$
12	**(a)** 30	Quantity × price is constant
13	**(d)** 20	$V = 2 + E - F$
14	**(c)** 40	Required total 180 minus 140
15	**(d)** $10x + 14$	Perimeter $= 2(l + b)$
16	**(c)** 156 m²	$(20+4)(15+4) - 20 \times 15$
17	**(a)** 0.3	$0.027 = 0.3^3$
18	**(a)** 1000	$a^m \times b^m = (ab)^m = (10)^3$
19	**(d)** 5	20 lies between 2^4 and 2^5
20	**(a)** 85°	$\angle B = \angle D$ and the angle sum is 360°

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$2x + 10 + 3x - 3$ (1); $5x + 7$ (1).	1+1
22	$l + b = 40$, so $b = 14$ cm (1); area $= 26 \times 14 = 364$ cm² (1).	1+1
23	$\frac{\sqrt[3]{27}}{\sqrt[3]{512}}$ (1) $= \frac{3}{8}$ (1).	1+1
24	$\frac{1}{32} = 2^{-5}$ (1); so $x = -5$ (1).	1+1
25	139 + 262 = 401 (1); CDI (1).	1+1
26	AO = OC gives $x = 4$ (1); AO $= 17$ cm so AC $= 34$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Decimal values 5 and 3 (1); product 15 (1); 1111_2 (1).	1+1+1
28	Sides $= 2k$ and $5k$; $2(2k + 5k) = 56$ (1); $k = 4$ (1); sides 8 cm and 20 cm (1).	1+1+1
29	Alternate sums for each trial; need difference 0 or 11 (1); $* = 0$ (1); check 8305 (1).	1+1+1
30	$197 = 200 - 3$ (1); $(200-3)^2 = 40000 - 1200 + 9$ (1); $= 38809$ (1).	1+1+1
31	A : B : C $= 3 : 6 : 5$ (1); one part $= 10$ (1); numbers 30, 60, 50 (1).	1+1+1
	OR: SI = ₹ 1800 (1); T = SI×100/(P×R) (1) = 3 years (1).	1+1+1
32	$18^2 + 24^2 = 900 = 30^2$ (1); right-angled with legs 18, 24 (1); area $= \frac{1}{2} \times 18 \times 24 = 216$ cm² (1).	1+1+1
	OR: A:B:C $= 3 : 6 : 8$ (1); one part $= 10$ (1); A = 30, B = 60, C = 80 (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	LCM(6, 8, 12) $= 24$ (3); required number $= 24 + 3 = 27$ (2).	3+2
34	(i) $103^2 = 10609$ (1.5); (ii) $88 \times 92 = 8096$ (1.5); (iii) $(76 + 24)(76 - 24) = 100 \times 52 = 5200$ (2).	1½+1½+2
35	Ratio $3 : 5$ (1); one part ₹ 100 (1); Anil ₹ 300 (1.5); Bhanu ₹ 500 (1.5).	1+1+1½+1½
36	Total sum = ₹ 10000 (1); R = SI×100/(sum×T) = 5% (2); interest on first = ₹ 400 (1); on second = ₹ 600 (1).	1+2+1+1
	OR: Height$^2 = 40^2 - 24^2 = 1024$, height $= 32$ cm (2); area $= \frac{1}{2} \times 48 \times 32 = 768$ cm² (2); perimeter $= 2 \times 40 + 48 = 128$ cm (1).	2+2+1
37	Work done in 8 days $= 8/20$ of the task; remaining $= \frac{3}{5}$ (1); workers now $= 18$; remaining man-days $= 144$ (2); days $= (12 \times 12) \div 18 = 8$ (2).	1+2+2
	OR: Stage 0: $3 \times 9 = 27$ cm (1); stage 1: $27 \times \frac{4}{3} = 36$ cm (2); stage 2: 48 cm (2).	1+2+2
38	Total $= 20 \times 50 + 30 \times 60 = 2800$ (2); mean $= 56$ (2); change $= 6$ (1).	2+2+1
	OR: $(10a + b) + (10b + a) = 11a + 11b = 11(a + b)$ (3); for 3 and 9: 132 (1); divisible by 11 (1).	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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